UNIVERSITATEA DIN BUCUREȘTI
FACULTATEA DE MATEMATICĂ ȘI INFORMATICĂ

TUTORIAT
CALCUL NUMERIC
IV

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Factorizarea QR

Exemplu:

$$Sa\ se\ scrie\ matricea\ A = \begin{pmatrix} 2 & 1 & 0 \\ 2 & 0 & 3 \\ 1 & 1 & 0 \end{pmatrix}\ sub\ forma\ A = Q * R, unde$$

$$Q\ este\ o\ matrice\ ortogonala\ respectiv\ R\ una\ superior\ triunghiulara.$$

Rezolvare:

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 2 & 0 & 3 \\ 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} a_{00} & a_{01} & a_{02} \\ a_{10} & a_{11} & a_{12} \\ a_{20} & a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} q_{00} & q_{01} & q_{02} \\ q_{10} & q_{11} & q_{12} \\ q_{20} & q_{21} & q_{22} \end{pmatrix} \begin{pmatrix} r_{00} & r_{01} & r_{02} \\ 0 & r_{11} & r_{12} \\ 0 & 0 & r_{22} \end{pmatrix}$$

$$Notam\ cu\ a_i, q_i, i = \overline{0,2}\ coloanele\ matricelor\ A\ respectiv\ Q$$

$$\begin{pmatrix} a_0 & a_1 & a_2 \end{pmatrix} = \begin{pmatrix} q_0 & q_1 & q_2 \end{pmatrix} \begin{pmatrix} r_{00} & r_{01} & r_{02} \\ 0 & r_{11} & r_{12} \\ 0 & 0 & r_{22} \end{pmatrix}$$

$$Rescriem\ sistemul:$$

$$\begin{cases} a_0 = q_0 r_{00} & (1) \\ a_1 = q_0 r_{01} + q_1 r_{11} & (2) \\ a_2 = q_0 r_{02} + q_1 r_{12} + q_2 r_{22} & (3) \end{cases}$$

$$Din\ ecuatia\ (1), pentru\ unicitate, alegem\ r_{00} = \|a_0\|_2 \Longrightarrow r_{00} = \sqrt{2^2 + 2^2 + 1^2} = 3$$

$$Obtinem\ direct: a_0 = 3q_0 \Longrightarrow q_0 = \frac{a_0}{3} \Longrightarrow \begin{cases} q_{00} = \dfrac{a_{00}}{3} \\ q_{10} = \dfrac{a_{10}}{3} \\ q_{20} = \dfrac{a_{20}}{3} \end{cases} \Longrightarrow \begin{cases} q_{00} = \dfrac{2}{3} \\ q_{10} = \dfrac{2}{3} \\ q_{20} = \dfrac{1}{3} \end{cases}$$

$$Prelucram\ ecuatia\ (2)\ pentru\ a\ obtine\ r_{01}; stim\ Q\ ortogonala\ deci\ putem\ inmulti$$

$$la\ stanga\ cu\ {q_0}^t:$$

$$Ecuatia\ (2)\ devine: {q_0}^t a_1 = {q_0}^t (q_0 r_{01} + q_1 r_{11})$$

$${q_0}^t a_1 = {q_0}^t q_0 r_{01} + {q_0}^t q_1 r_{11};$$

$$dar\ {q_0}^t q_0 = 1, {q_0}^t q_1 = 0\ (Q - ortogonala) \Longrightarrow$$

$$\Rightarrow {q_0}^t a_1 = r_{01} \Rightarrow r_{01} = \begin{pmatrix} \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \frac{2}{3} + 0 + \frac{1}{3} = \frac{3}{3} = 1$$

Revenim in (2):

$$a_1 = q_0 r_{01} + q_1 r_{11} \Rightarrow q_1 r_{11} = a_1 - q_0 r_{01} \Rightarrow q_1 r_{11} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - 1 * \begin{pmatrix} \frac{2}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{pmatrix} = \begin{pmatrix} \frac{1}{3} \\ -\frac{2}{3} \\ \frac{2}{3} \end{pmatrix}$$

Alegem r_{11} *norma euclidiana a vectorului adineauri obtinut* $\Rightarrow \cdots \Rightarrow r_{11} = 1 \Rightarrow$

$$\Rightarrow q_1 = \begin{pmatrix} \frac{1}{3} \\ -\frac{2}{3} \\ \frac{2}{3} \end{pmatrix} \Rightarrow \begin{cases} q_{01} = \frac{1}{3} \\ q_{11} = -\frac{2}{3} \\ q_{12} = \frac{2}{3} \end{cases}$$

Procedam analog operatiilor anterioare si pentru ecuatia (3), *dorim sa aflam*

r_{02} *respectiv* r_{12}.

Intai inmultim la stanga cu ${q_0}^t$:

$$Ecuatia\ (3)\ devine: {q_0}^t a_2 = {q_0}^t (q_0 r_{02} + q_1 r_{12} + q_2 r_{22})$$

$${q_0}^t a_2 = {q_0}^t q_0 r_{02} + {q_0}^t q_1 r_{12} + {q_0}^t q_2 r_{22};$$

$$dar\ {q_0}^t q_0 = 1, {q_0}^t q_1 = 0, {q_0}^t q_2 = 0\ (Q - ortogonala) \Rightarrow$$

$$\Rightarrow r_{02} = {q_0}^t a_2 = \begin{pmatrix} \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix} = 2$$

Apoi inmultim la stanga aceeasi ecuatie initiala (3) *cu* ${q_1}^t$:

$$Ecuatia\ (3)\ devine: {q_1}^t a_2 = {q_1}^t (q_0 r_{02} + q_1 r_{12} + q_2 r_{22})$$

$${q_1}^t a_2 = {q_1}^t q_0 r_{02} + {q_1}^t q_1 r_{12} + {q_1}^t q_2 r_{22};$$

$$dar\ {q_1}^t q_0 = 0, {q_1}^t q_1 = 1, {q_1}^t q_2 = 0\ (Q - ortogonala) \Rightarrow$$

$$\Rightarrow r_{12} = {q_1}^t a_2 = \begin{pmatrix} \frac{1}{3} & -\frac{2}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix} = -2$$

$Revenim\ in\ (3)$:

$$a_2 = q_0 r_{02} + q_1 r_{12} + q_2 r_{22} \Longrightarrow q_2 r_{22} = a_2 - q_0 r_{02} - q_1 r_{12} =$$

$$= \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix} - 2 * \begin{pmatrix} \frac{2}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{pmatrix} - (-2) * \begin{pmatrix} \frac{1}{3} \\ -\frac{2}{3} \\ \frac{2}{3} \end{pmatrix} = \cdots = \begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \\ \frac{2}{3} \end{pmatrix}$$

$$Alegem\ r_{22}\ norma\ euclidiana\ a\ \ vectorului\ \Longrightarrow\ \ldots \Longrightarrow r_{11} = 1 \Longrightarrow$$

$$\Longrightarrow\ q_2 = \begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \\ \frac{2}{3} \end{pmatrix} \Longrightarrow \begin{cases} q_{02} = -\frac{2}{3} \\ q_{12} = \frac{1}{3} \\ q_{22} = \frac{2}{3} \end{cases}$$

$$Am\ obtinut:$$

$$Q = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \end{pmatrix}\ si\ respectiv\ R = \begin{pmatrix} 3 & 1 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}\ Q.E.D$$

Exerciții propuse:

$$Sa\ se\ rezolve\ sistemul\ de\ ecuatii \begin{cases} 2x_0 - x_2 = 1 \\ 2x_0 + 3x_1 + 2x_2 = 10 \\ x_0 + 3x_1 + 7x_2 = 14 \end{cases}\ folosind\ factorizarea\ QR.$$

$$Q * R *\ x = b$$

$$Q^t * Q * R * x =\ Q^t * b$$

$$I_3 * R * x =\ Q^t * b \Longrightarrow x = \cdots$$